

CPLREADER - A CFD DATA READER AND ACOUSTIC S

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2 Usage

3 Description

In computational aero-acoustics the so-called hybrid methods are quite common. These methods try to extract source terms for acoustics from CFD data. There exists a wide variety of such methods. The volume based approach used by *CFS++*¹ [11] is the computation of Lighthill source terms in a FEM framework directly on the CFD mesh. Since the CFD mesh is usually too fine to perform an acoustic computation on it, the source terms have to be conservatively interpolated to a dedicated acoustic grid. The first step is the job of *cplreader*. The interpolation is performed within *CFS++*.

The

4 Options

Table 1: Command line options of *cp/reader*.

Parameter	Default Value	Short Description
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Table 1 – continued from previous page

Parameter	Default Value	Short Description
		This parameter specifies the factors to be multiplied with the corresponding components of the velocity field. This can be useful, if the CFD computation used a different scaling (cf. OpenFOAM simulations of the Freiberg guys)
--density	1.0	Density of the fluid. Default density is 1.0 for testing. The mean density of the fluid may be specified by this parameter. Some practically relevant values include: air 1.2041kg/m^3 (at

5 Theory

$$\begin{aligned}
 L_{ij} &= \frac{1}{ij} \frac{0}{ij} \\
 &= v_i v_j + (p - p_0) c_0^2 (\quad 0)_{ij} \frac{0}{ij} ;
 \end{aligned}
 \tag{12}$$

with v^B

Element type	Reference points				Nodal basis functions
	Node i				
Triangle	1	0	0	0	$N_1 = 1$
	2	1	0	0	$N_2 =$
	3	0	1	0	$N_3 =$

Quadrilateral

The time discretization is performed by applying a standard Newmark algorithm, which results in the following time-stepping scheme (effective mass matrix formulation) [10]:

$$\left(\frac{J}{\Delta t} + \frac{F}{3} \right) \mathbf{u}^{n+1} = \left(\frac{J}{\Delta t} - \frac{F}{3} \right) \mathbf{u}^n + \frac{F}{3} \mathbf{u}^{n-1} + \frac{\Delta t^2}{6} \mathbf{F}^{n+1/2}$$

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6 CFD Data Readers

6.1 FASTEST-3D

6.1.1 Required File Layout

```
./name  
|-- name_01.coord  
|-- name_01.topo  
|-- name_01_0001.dat  
|-- name_01_0002.dat  
|-- name_02.coord  
|-- name_02.topo  
|-- name_02_0001.dat  
'-- name_02_0002.dat
```

6.1.2 Special Options

6.1.3 ASCII File Format

.coord-File Format

```
1 1 2 6  
0. 0. 0.0  
0. 0.03125 0.0  
0. 0.09375 0.0  
0.03125 0. 0.0  
0.03125 0.03125 0.0  
0.03125 0.09375 0.0
```

.topo-File Format

```
1 1 2 2 4  
4 5 2 1  
5 6 3 2
```

.dat-File Format

```
1  
-5.30794066E-06  
-1.39465176E-05  
-2.06977642E-05  
-1.94136509E-05  
-3.94903112E-05  
-7.02913899E-05
```

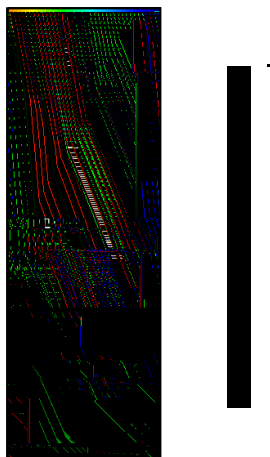


Figure 5: Visualization of dataset defined by above files.

6.1.4 Examples

6.2 ANSYS CFX

6.2.1 Required File Layout

[Faces]

7 Other Readers and Mesh Generators

7.1 ANSYS Mesh Converter

7.2 Mesh Generator for Spherical Shells

8 Copyright

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[16] C.K.W. Tam. Computational aeroacoustics. *AIAA Journal*, 33:1788–1796, 1995.

[17] C.K.W. Tam. Supersonic jet noise. *Annual review of fluid mechanics*, 27:17–43, 1995.